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Sumaiya Sande

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Sandipan Dey

sandipan.dey@gmail.com



Sandipan Dey

geekforeverumbc@gmail.com

Pros and Cons of popular Supervised Learning Algorithms

LASSORidge
LDA NaiveBayes
Logistic KNN SVM
XGBoost RandomForest

We all have used one of the following supervised learning algorithms for predictive analysis:

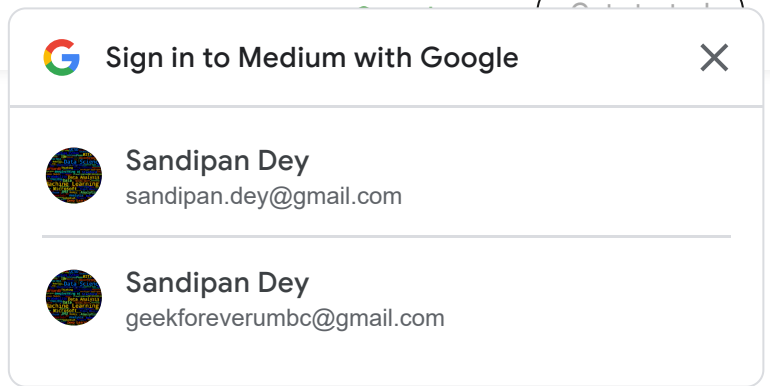
1. Logistic Regression
2. Ridge Regression
3. LASSO Regression
4. Linear Discriminant Analysis (LDA)
5. K Nearest Neighbors (KNN)
6. Naive Bayes (NB)





9. Random Forest (RF)

10. Gradient Boosting



But have you thought of their pros or cons? Here I have listed few :

1. Logistic Regression :

Pros :

- a) It is used when the data is linearly separable.
- b) It is easier to implement, interpret and very efficient to train.
- c) It gives the measure of how importance of a predictor both in positive or negative direction.

Cons :

- a) It can overfit in high dimensional datasets.
- b) Does not support non-linear relationship between the predictor and the outcome.

2. Ridge Regression :

Pros :

- a) Prevents over-fitting in higher dimensions.
- b) Balances Bias-variance trade-off. Sometimes having higher bias than zero can give better fit than high variance and zero bias.

Cons :





b) We need to select optimal alpha (lambda)

c) Model interpretability is low.

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Sandipan Dey
sandipan.dey@gmail.com



Sandipan Dey
geekforeverumbc@gmail.com

3. LASSO Regression :

Pros :

a) Performs feature selection by shrinking coefficients towards zero.

b) Avoids over fitting.

Cons :

a) Selected features can be highly biased.

b) For $n \ll p$ (n-number of data points, p-number of features), LASSO selects at most n features.

c) For different bootstrapped data, the features selected can be very different.

4. Linear Discriminant Analysis (LDA) :

Pros :

a) It is simple, fast and portable algorithm. It still beats some algorithms (logistic regression) when its assumptions are met

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Cons :

a) It requires normal distribution assumption on features/predictors.

b) Sometimes not good for few categories variables.





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Sandipan Dey

sandipan.dey@gmail.com



Sandipan Dey

geekforeverumbc@gmail.com

a) It is the most simple algorithm to find the k nearest neighbors k .

b) One can plug in any distance metric for working with complex objects, like text and basically anything you can define distance metric for.

c) This algorithm can be used for classification, ranking, regression (using neighbors average or weighted average), recommendations, missing value imputation etc.

Cons :

a) KNN is a lazy learner because it doesn't learn a model weights or function from the training data but "memorizes" the training dataset instead. Hence, it takes longer time for inference than training.

b) It is a distance based-approach hence the model can be badly affected by outliers, in other words, it's prone to overfitting.

c) Model size grows with the new data incorporated.

d) The model suffers from the curse of dimensionality.

6. Naive Bayes (NB)

Pros :

a) It is easy and fast to predict class of test data set. It also performs well in multi class prediction.

b) When assumption of independence holds, a NB classifier performs better compared to other models like logistic regression and you need less training data.

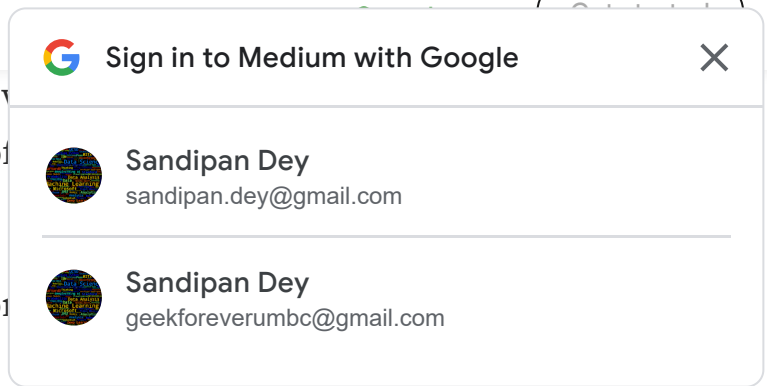




make a prediction. This is often known as the smoothing technique. One of the methods used is Laplace estimation.

b) There is a strong set of assumptions: normal, multinomial etc.

c) There is also the assumption of independence in predictors. In real life, it is almost impossible that we get predictors which are completely independent.



7. Support Vector Machine (SVM) :

Pros :

a) It works really well with a clear margin of separation

b) It is effective in high dimensional spaces.

c) It uses a subset of training points in the decision function (called support vectors), so it is also memory efficient.

d) It is also efficient in non-linear model fitting.

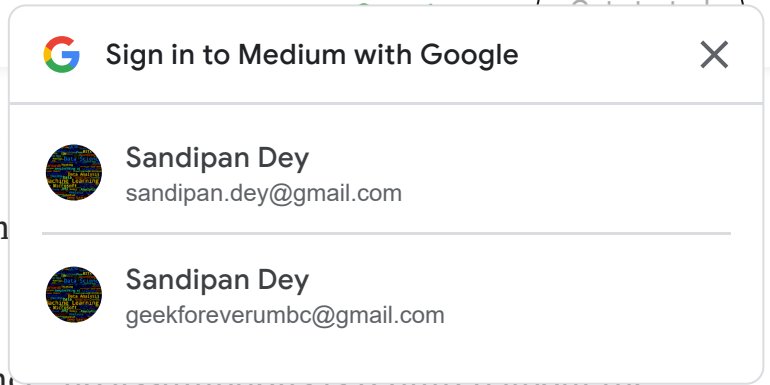
Cons :

a) It doesn't perform well when we have large data set because the training becomes time-consuming.

b) It also doesn't perform very well, when the data set has more noise i.e. target classes are overlapping.

c) SVM doesn't directly output probability. Other methods need to be used to convert the output from SVM to probability.





- a) Easy to understand and interpret,
- b) It requires little data preprocessing, standardization and so on.
- c) It is a non-parametric model. Hence, no assumptions is required about the distribution of the data.
- d) Feature selection happens automatically. So, unimportant features will not influence the result.

Cons :

- a) It tends to overfit.
- b) It is very sensitive . Small change in the data can affect prediction greatly (High variance).

9. Random Forest (RF) :

Pros :

- a) It is robust to outliers. It lowers risk of overfitting.
- b) It also works well with non-linear data.
- c) It runs efficiently on a large dataset.
- d) Usually it gives better accuracy than other classification algorithms.

Cons :

- a) Random forests are found to be biased while dealing with categorical variables.
- b) Slow Training.





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Sandipan Dey

sandipan.dey@gmail.com



Sandipan Dey

geekforeverumbc@gmail.com

10. Gradient Boosting :

Pros :

a) Boosting comes with an easy to re prediction interpretations easy to ha

b) Boosting is a resilient method that curbs over-fitting easily.

Cons :

a) It is sensitive to outliers since every classifier is obliged to fix the errors in the predecessors. Thus, the method is too dependent on outliers.

b) Boosting is almost impossible to scale up. This is because every estimator bases its correctness on the previous predictors, thus making the procedure difficult to streamline.

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Sandipan Dey

sandipan.dey@gmail.com



Sandipan Dey

geekforeverumbc@gmail.com

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